Infectious Disease Models in Crisis Decision-Making: Roles, Pathways, and Realities in the Policy Arena – 12 December 2025 – Q&A

A. Communicating Uncertainty

1. *How should we address and communicate about uncertainty in analyses?*
2. *Dealing with model uncertainty in policy making and how to communicate this to various stakeholders and the general population without losing trust.*
3. *How important is the level of uncertainty in a model or forecast for shaping the final decision or policy?*

Most of these questions were discussed during the workshop. The most important points are reiterated below, as an answer to the question "Uncertainty is a given in modelling. How do policy makers handle this uncertainty in their decision making?"

Uncertainty is the name of the game. However, it is always best to explain the uncertainty (nature and impact). It is helpful to tailor communication about uncertainty to the ones who must make or understand the decisions (politicians and/or public). How uncertainty is weighed is partially an individual preference. This is why communication about modelling is always part of a dialogue.

B. Effective Model Communication

4. *In what form is communication about infectious disease models most effective (level of detail/technicality, report or slide deck or some other form, figures or tables)?*

This depends on timing and context of the request. Outside of pandemics/high-level outbreaks, there is often more time to discuss based on (lengthy) reports. As always, structured slide decks will often help the discussions during meetings and are a method to communicate fast and effectively. During outbreaks, time is of the essence and methods of communication that are effective under time constraints become more important, such as executive summaries (bullet point lists) of main findings, conclusions, and sources of uncertainty, or in the form of direct replies to policy questions. As mentioned during the workshop, informed advice also requires an ongoing dialogue and going back and forth between scientists and policy makers to develop a more fine-grained understanding of policy problems and interventions considered (by policymakers or scientists).

C. AI

5. *In healthcare AI algorithms are already being used for diagnosis, and there are a lot of developments with AI for predictions of recovery and disease progress.*
6. *In what way is AI-powered modelling for infectious diseases modelling acceptable for the Ministry?*

During the workshop the question "in what way is AI-powered modelling for infectious diseases modelling acceptable for VWS" was discussed. Several factors are important to consider here. Decisions need to be explainable to the public due to procedural justice considerations. That implies that we need to consider whether AI models can withstand scrutiny and are transparent—are they acceptable to the public? It's also important that the model is scientifically validated within your own field (modelling), and that you can personally explain it before presenting it to the public. Most of these discussions need to happen within scientific communities. It tends to help if models withstand academic scrutiny in general.

Moreover, whether AI can be used in policy making in general is currently under consideration at the Ministry of Domestic Affairs. This will take some time. We, Ministry of Health, Welfare and Sport, are generally interested in AI in modelling as a way of progressing the field of modelling and look forward to interacting with the modelling community on this and are already doing so through funding, being a stakeholder in research projects, and/or supervising policy trainees.

D. Scientist-Policymaker Collaboration

7. *Where do you see the tension in infectious disease modellers and policy making? And which aspects do you think modellers could improve upon in interacting with policy makers?*

Infectious Disease Models in Crisis Decision-Making: Roles, Pathways, and Realities in the Policy Arena – 12 December 2025 – Q&A

As discussed during the workshop, tensions often arise on the axis of scientific accuracy and the need for concrete input in the policy-making process in heavily limited timeframes. The need for concrete input does not always align with the nuance of scientific accuracy (e.g., models cannot predict the future and reality is not mechanical). Moreover, what is academically interesting may not be always interesting from a policy perspective. Pipelining knowledge agendas and being stakeholders in each other's process will help in gaining a better understanding of the requirements and desired output on both sides, as well as interacting between policy makers and scientists on a regular basis will help to shape more effective engagement when needs are there.

8. *Do VWS and the RIVM have a plan to engage modelling and data-analysis experts, at least as a sounding board and ideally as collaborative partners, should a new epidemic crisis erupt? This requires careful consideration of financial and confidentiality issues, which should not be postponed until a new crisis is imminent.*

Both VWS and RIVM acknowledge the added value of involving experts from the Collaborative Platform in modelling and data analysis for policy advice during a new crisis. This requires an organizational structure, supported by legal agreements and a technical infrastructure for exchange of code and data. Development and maintenance of such an organization is one of the objectives of the Collaborative Platform and will be done in collaboration with all partners in the core group.

E. Policy Decision-Making Processes

9. *How do you decide on potential policies during an infectious disease outbreak? For instance, how do you decide on measures like digital contact tracing, admission passes? To what extent is effectiveness decisive and to what extent are interventions not possible because the public/politicians are against them.*

Much of this was already discussed during the workshop and is linked to the crisis decision-making structure that is formalized at this moment. Some measures are explicitly formalized in the Public Health Act (see paragraph 8, article 58a-za for more details). Decisions should always follow the logic decision-making structure, but the nature of the policies considered will depend on the nature of the pathogen, the epidemiological situation, and legal boundaries. This will also depend on the scale of the outbreak, as potential policies during pandemics are set by multiple departments, changing the dynamics of the decision-making process.

10. *How much autonomy one can expect from local authorities? Will all decisions e.g., about lockdown measures be taken on a national level?*

Local autonomy is formalized in the Public Health Act. Local governors (mayors/chairs of safety regions) take decisions on individual measures. Collective measures during a pandemic (group A1 infectious disease) are decided upon by the cabinet. In line with article 58e, local governors can make certain exceptions to collective measures (such as closure of specific public spaces). For a lengthier discussion, we would like to point to the 'memorie van toelichting' of the first tranche update of the Public Health Act: https://www.tweedekamer.nl/downloads/document?id=2022D35900

11. *What is the role of quantitative input/predictions in the decision-making process?*

This was discussed during the workshop, quantitative input (for policy) is mostly used to draft advice from the outbreak management team/response team at the RIVM.

F. Use and Selection of Evidence

12. *How do you weigh scientific evidence-based advice coming from non-RIVM sources? What drives the decision to take some expert comments into account and others not?*

We view the RIVM as 'trusted advisor', given the legal obligations the RIVM has (as stipulated in both the 'Law on the RIVM' and the Public Health Act). However, the RIVM also has an important

role in summarizing and interpreting scientific evidence from non-RIVM sources. This allows the Ministry of Health. Welfare and Sport to stay up to date on scientific evidence in general. We also fund academic research through intermediaries such as ZonMw and TNO and tend to explicitly appreciate the academic output of such projects, both internally and externally (to parliament).

13. *In how far do political arguments play a role when asking advice from modellers? With political arguments I mean for instance: delaying decisions; or seeking additional arguments for a decision already taken; or attracting attention to a particular topic.*

Political arguments play a role in the sense that advice questions are generally asked based on policy preferences. That is, advice is *generally* not asked regarding policy preferences that have no chance of being implemented (due to legal, political, which includes economical, or societal reasons). However, RIVM may and does also give unsolicited advice (e.g., on other options than those asked advice on explicitly, if deemed relevant in terms of infection control).

G. Engaging with Policymakers and Offering Expertise

14. *How do we know which questions are relevant to policy makers, and what the deadlines for decision making are?*

Deadlines for decision-making are mostly formalized in law (see again paragraph 8 of the Public Health Act) during pandemics. Outside of a pandemic, there are no formal guidelines on decision-making during outbreaks. Deadlines are usually very short for modelling input, given the nature of crises.

Relevant questions for policy makers concern all questions regarding situational awareness (what the epidemic currently looks like). It seems trivial, but it should be noted that situational awareness is highly relevant to the public as well. Furthermore, short-term projections and (relative) effectiveness of interventions are also relevant for policy makers to guide decision making, besides situational awareness.

15. *If we would like to offer my advice or share expertise, who should we contact? Context: work at a university, not at the RIVM. Should we contact the media, the KNAW, the ministry of OCW, the ministry of VWS, the OMT, the RIVM?*

Outside of pandemics (most of the time, luckily), the ministry is quite open to discussions with academics through various means (workshops, conferences which we attend, ZonMw project meetings, etc.). During a pandemic, time is often more limited, and we will become more selective due to time constraints. Coordinating expertise with the RIVM is generally advised. As indicated in the answer to question 8 above, an objective of the Collaborative Platform is to create a more formal structure to contribute to modelling and data analysis for policy advice.

H: Modelling and Preparedness for Intentional Pathogen Release

16. *Should the case of intentionally released pathogens (e.g., bioterrorism, biological warfare) be treated and modelled differently than a pandemic, and how should the collaboration with other ministries (e.g., defense) be to be prepared for this?*

Whether the substantive/mechanistic part of modelling changes depending on the nature of the pathogen is a question for the scientific community. However, be aware that during such an event the nature of decision-making and communication will change rapidly. First, the influence of the Ministry of Health, Welfare and Sport will change as the Ministry of Justice/Defense and the National Coordinator for Security and Counterterrorism (NCTV) will likely take a bigger role. Decisions are made much more quickly and space for communicating about uncertainty will most likely also change.

Infectious Disease Models in Crisis Decision-Making: Roles, Pathways, and Realities in the Policy Arena – 12 December 2025 – Q&A

In general, it's becoming more important to consider this scenario and academically there's probably also more interest in this. We are working on scenario exercises here and are also keen to learn more on this topic.